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RAMAIAH
Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END EXAMINATIONS – APRIL 2023

Program	:	**B.E. – Information Science and Engineering**	**Semester**	:	**III**
Course Name	:	**Object Oriented Programming with Java**	**Max. Marks**	:	**100**
Course Code	:	**IS34 / IS35B**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1.	a)	If local variable has the same name as an instance variable, how Java handles this situation? Illustrate the same with code snippet.	CO1	(07)
	b)	"Class Methods and instance variables can be accessed directly in another class without creating Objects" Justify the same with code snippet.	CO1	(07)
	c)	Write a Java program to i. get the character at the given index within the String. ii. compare two strings. iii. create a string array with initialization and prints the array elements on the standard output.	CO1	(06)
2.	a)	Differentiate between Static and Non-Static Method with code snippet.	CO1	(07)
	b)	Demonstrate Overloading of Varargs(Variable arguments) Method with the example code.	CO1	(07)
	c)	Write a Java Program which accepts command line arguments and prints the same on the standard output.	CO1	(06)

UNIT - II

3.	a)	Create a class Tax with abstract method calTax(double amount). Create its two sub classes USTax and CanadaTax. Write a program that calculate s appropriate tax based on whether USTax object or CanadaTax object is created.	CO2	(08)
	b)	What is dynamic method dispatch? Illustrate an example using Java code.	CO2	(06)
	c)	List any six methods of Object class and state their purpose.	CO2	(06)
4.	a)	Write a program to implement stack of integers by implementing an interface which has push() and pop() methods declaration.	CO2	(08)
	b)	List four access modifiers in Java and illustrate their scope through Java code.	CO2	(06)
	c)	List four access modifiers and tabulate their scope with respect to class, subclass and package.	CO2	(06)

UNIT - III

5.	a)	Describe the exception handling keywords with syntax.	CO3	(08)
	b)	Write a Java program to read 10 bank transactions and store in an array of integers. Create nested try blocks to such that the inner try block handles ArrayIndexOutOfBoundsException while the outer try block handles the ArithemeticException (division by zero).	CO3	(08)
	c)	List the constructors and methods of Throwable class which support chained exceptions in java along with its argument description.	CO3	(04)
6.	a)	Demonstrate the exception handling mechanism with an example.	CO3	(08)
	b)	Identify the purpose of following exceptions. Write a Java code snippet to demonstrate the usage of the same. i) ArrayOutOfBoundException ii) NumberFormatException	CO3	(08)
	c)	Describe the need for user-defined exceptions. Write the syntax to create user-defined exceptions in Java.	CO3	(04)

UNIT- IV

7.	a)	Explain the working of following methods in relation to wrapper objects: valueOf(), parseInt(String str, radix r), intValue(), toString().	CO4	(08)
	b)	What is auto-boxing and auto-unboxing? Write Java code which demonstrates the same.	CO4	(06)
	c)	List three methods of Iterator object and write Java code that enumeratestheir working.	CO4	(06)
8.	a)	List and give example code of usage for any four List interface specific methods.	CO4	(08)
	b)	Differentiate between: i. ArrayList and LinkedList classes ii. List and Set Interfaces.	CO4	(06)
	c)	Implement Queue of String objects using methods present only in LinkedList class.	CO4	(06)

UNIT - V

9.	a)	"When two or more threads need access to a shared resource, they need some way to ensure that the resource will be used by only one thread at a time" , What mechanism java uses to address the same? Illustrate with an example code.	CO5	(08)
	b)	Write a java code to demonstrate block lambda which computes and return the factorial of an int value.	CO5	(07)
	c)	Illustrate how current state of a thread can be obtained with an example.	CO5	(05)
10.	a)	Demonstrate the creation of multiple threads with an example code.	CO5	(08)
	b)	Define Lambda Expression with an example.	CO5	(05)
	c)	Write a java code to throw an exception from a lambda expression.	CO5	(07)

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